

BBA 2421 BUSINESS MATHEMATICS

LECTURE 6

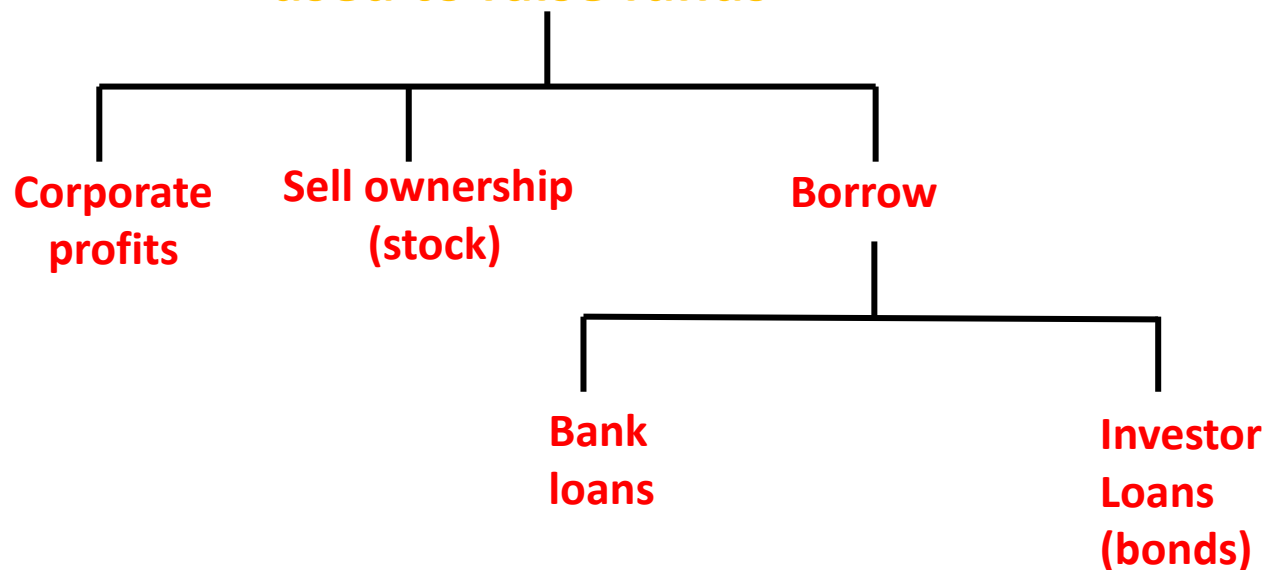
BONDS, STOCKS AND SINKING FUNDS



Introduction

- Corporations need money for many purposes, including to make payroll or pay for a new building or equipment.

Methods managers used to raise funds



Bonds

- **Bonds** are loans to firms, consisting of legally binding contracts between the lenders and the borrowers.
- Interest payments are made every year, and repayment at a specific date in the future.
- Bondholders do not own part of the corporation.
- Countries, cities, and even churches also borrow money using bonds.
- The prices of bonds vary according to the prevailing interest rate.



Terms associated with bonds

- **Coupon rate** – The interest rate stated on the bond.
- **Face value** – The amount to be repaid when the loan matures. Face value is sometimes referred to as maturity value or par value.
- **Interest** – Interest is found using the face value of the bond, the coupon rate, and the compounding period. Most bonds require interest to be paid every six months.
- **Issuer** – The issuer is the corporation or organization that is making the loan (selling the bond).
- **Maturity date** – Date at which the bond matures and the issuer repays the face value to the current owner of the bond.
- **Term** – The length of time until the bond matures. Terms on individual bonds are determined before the bond is issued, but the terms are often in the range of 1 to 30 years



- A company is said to go bankrupt if it can no longer meet its financial obligations to suppliers, banks, bondholders, and others.
- Remaining assets are then used to pay off debt, including bonds.
- Bondholders often receive only 10 to 50 cents on every dollar originally loaned.
- Corporations frequently use substantial amounts of debt to build factories, expand operations, or buy other companies.
- The interest that must be paid on that debt is the cost of having debt.
- The larger the debt, the greater the amount of revenue the company must set aside to pay interest.



Read bond tables

- The face value, or par value, of a bond is the original amount of money borrowed by a company. Most public corporations issue bonds with a par value of $K1000$
- Principal and any interest due must be paid when a bond matures.
 - But bonds can be sold through a bond dealer, even before they mature.
- The value of a bond is strongly influenced by the **credit history of the firm** and **market conditions** at the time of the sale.
- Market interest rates fluctuate from year to year, yet any one bond pays exactly the same dollar amount of interest each year.
- The price of a bond fluctuates in the opposite direction of interest rates.



Example

Kafue was in an automobile accident that put her in the hospital for 3 weeks and required months of rehabilitation. The other driver was at fault, and his insurance company paid Barrett the liability limits on his policy of *K*50,000. Barrett needs monthly income and is thinking about investing the funds in Wells Fargo (WFC) bonds that mature in 2046. Analyze the data in the table.

Company (Ticker)	Coupon	Maturity	Last price	Last yield	Est \$ Vol (000's)
Wells Fargo (WFC)	4.75	7 December, 2046	99.004	4.813	171,626



Solution

- a) The ticker symbol for Wells Fargo is WFC.
- b) Annual interest paid per bond = 4.75% of $K1000$ face value = $K47.50$.
- c) The bond matures on December 7, 2046, when Wells Fargo must repay the $K1000$ face value to the owner of the bond.
- d) The last price at which the bond traded for the day was 99.004% of the face value of $K1000$. Last price = $0.99004 \times K1000 = K990.04$
- e) An investor holding the bond until maturity in 2046 would earn 4.813% per year on the investment. This is often called the yield to maturity.
- f) The estimated face value of all the Wells Fargo bonds that sold during the week was

$$Volume = 171,626 \times K1000 = K171,626,000$$



Bond table

Company (Ticker)	Coupon	Maturity	Last price	Last yield	Est <i>K</i> Vol (000's)
Oracle (ORCL)	1.900	15 Sep, 2021	96.953	2.587	157,667
Apple (AAPL)	3.850	04 Aug, 2046	93.292	4.250	158,021
Ford Motor Company (F)	5.291	08 Dec, 2046	98.996	5.359	219,115
Kraft Heinz Foods Co (KHC)	3.000	01 June, 2026	92.488	3.961	158690
Viacom (VIA)	4.375	15 Mar, 2043	78.817	5.985	310,905
Citigroup (C)	2.900	08 Dec, 2021	99.095	3.098	196,479
Viacom (VIA)	3.450	04 Oct, 2026	90.983	4.604	317,916
Citigroup (C)	3.2	21 Oct, 2026	94.411	3.889	309.316
Hsbc Holdings (HSBC)	4.375	23 Nov, 2026	99.147	4.482	159,976



Example

- Using the bond table find the estimated volume sold during the prior week and the last sale price for the following bonds.
 - a. Oracle (ORCL) maturing in 2021
 - b. Apple (AAPL) maturing in 2046
 - c. Ford Motor Company (F) maturing in 2046



Solution

Company	Volume Sold	Last Sale
		Price per Bond
a. Oracle	<i>K</i> 157,638,000	<i>K</i> 969.53
b. Apple	<i>K</i> 158,021,000	<i>K</i> 932.92
c. Ford Motor Co.	<i>K</i> 219,115,000	<i>K</i> 989.96



Find the commission charge on bonds and the cost of bonds 12

- A common charge to buy or sell a few bonds is $K1$ per bond, but commissions are much lower for large transactions.
- Commission costs must be taken into account when calculating the effective interest rate of a bond.
- **Effective interest rate** is calculated by dividing the total interest the investor earns by the total cost of the bond including commissions.
- **Last yield** takes into consideration the maturity value of the bond, the time to maturity, and all interest payments, but not commissions.



Example

Assume that the sales charge is $K1$ per bond, and find the following for Kraft Heinz Foods Co (KHC) maturing in 2026.

- a) The total cost of purchasing 20 bonds
- b) The total annual interest paid on these bonds
- c) The effective interest rate to the buyer, including the cost of buying the bonds



Solution

(a) Total cost = (Price per bond + Sales charge per bond) × (Number of bonds)

$$= (K924.88 + K1) \times 20 = K18,517.60$$

(b) Annual interest = Coupon rate × Par value of bond × Number of bonds

$$= (0.03 \times K1000) \times 20 \text{ bonds} = K600$$

$$(c) \text{ Effective rate} = \frac{\text{Total interest}}{\text{Total cost of bonds}} = \frac{K600}{K18517.60} = 3.2\%$$



Finding the Net Amount from the Sale of Bonds

Example

Find the amount received from the sale of 50 Viacom (VIA) bonds maturing in 2043. Assume a commission of $K1$ per bond.

Solution

Amount received = (Sales price of a bond - Sales charge per bond) $\times$ Number of bonds

$$= (K788.17 - K1) \times 50 = K39,358.50$$



Stocks

- Many businesses are set up as **corporations**.
- **Publicly held corporations** are those owned by the public; their stocks are traded daily in markets called stock markets.
- **Privately held corporations** are owned by one or a few individuals, and their stock is not traded on a market
- A **corporation** is a form of business that gives the owners (the stockholders) limited liability
- A corporation is set up with money, or capital, raised through the sale of **shares of stock**.
- **A share of stock** represents partial ownership of a corporation.



Types of stocks

- The two types of stock are **preferred stock** and **common stock**.
- Preferred stock-have certain rights over common stockholders.
- For example, owners of preferred stock must be paid dividends before any dividends can be paid to owners of common stock.
- Also, corporate debt and preferred shareholders must be paid before common shareholders receive anything in the event that a corporation declares bankruptcy.



- The shares of **publicly held corporations** are typically owned by many different individuals and institutions.
- Share prices of these firms are determined by supply and demand in public markets called **stock exchanges**
- Businesses and individuals buy stock to make money.
- Stock prices usually go up over time, but not always since recessions drive stock prices lower and companies can go bankrupt.
- Many, but not all, publicly held corporations pay dividends—regular payments to stockholders usually made every quarter.
- Individual stocks can be purchased using **stockbrokers** or individuals who make their living by advising investors.
- Stocks can also be purchased directly by individuals through any of the numerous **discount brokers**.



Reading the stock table

Example

									Earnings			
52-Wk		Name	Tick Sym	Vol. 100s	Yld.	P/E	Week's		Latest Year	This Year	Next Year	Div Amt.
High	Low						Last	Chg.				
49.40	25.31	BestBuy	BBY	242901	2.4	14	47.00	-2.00	2.56	3.26	3.47	0.28

After receiving his first paycheck, Darnell Johnson went to Best Buy to look at iPhones and MP3 players. He liked the company so well that he decided to do some research on it. Analyze the information about Best Buy stock given in the table.



Solution

- a) The highest price at which the stock sold during the year was *K*49.40.
- b) The lowest price at which the stock sold during the year was *K*25.31.
- c) The ticker symbol under which the stock trades on the market is BBY.
- d) The volume of shares sold during the day was $100 \times 242901 = 24,290,100$ shares.
- e) The annual dividend yield is 2.4% of the current stock price.
- f) The ratio of stock price to earnings was 14



Solution continued...

- g) The stock closed at $K47.00$ per share at the end of the week.
- h) The price of the stock was down $K2.00$ compared to the close at the end of the prior week.
- i) Best Buy earned $K2.56$ per share last year. It is expected to earn $K3.26$ per share this year and $K3.47$ per share next year.
- j) The most recent dividend was $K0.28$ per share. We know it is a quarterly dividend, since total annual dividend ($K0.28 \times 4 = K1.12$) equates to the 2.4% annual dividend yield noted in (e) above.



Finding the Cost of Stocks

Ignoring commissions, find the cost for each of the following purchases.

- (a) 500 shares of Bank of America (BAC) at the close for the week
- (b) 800 shares of Bladex (BLX) at the low for the year
- (c) Then find the combined annual dividend from these two investments.



Solution

- a. $500 \text{ shares} \times K22.66 \text{ per share} = K11,330$
- b. $800 \text{ shares} \times K19.63 \text{ per share} = K15,704$
- c. Quarterly dividend from Bank of America: $K0.075 \times 500 = K 37.50$
 Quarterly dividend from Bladex: $K0.385 \times 800 = \underline{K308.00}$
 Total quarterly dividend = $K345.50$
 Total annual dividend = $K345.50 \times 4 \text{ quarters in a year} = K1382.$



- There is no certain way of choosing, stocks that will increase in price.
- Two stock ratios that people commonly look at before buying shares of a company are the **current yield** and the **price–earnings ratio**.
- It is used to compare the dividends paid by stocks selling at different prices.

$$\text{Current yield} = \frac{\text{Annual dividend per share}}{\text{Closing price per share}}$$

$$\text{Price – eanings}(PE)\text{ratio} = \frac{\text{Closing price per share}}{\text{Annual income per share}}$$



Example

Use data in the stock table to estimate the current yield for Bankrate (RATE) and Barclays (BCS) rounded to the nearest tenth of a percent.

Solution

$$\text{Current yield for Bankrate} = \frac{\text{Annual dividend}}{\text{Closing price}} = \frac{K0}{K10.35} = 0\%$$

$$\text{Current yield for Barclays} = \frac{\text{Annual dividend}}{\text{Closing price}} = \frac{K0.0528 \text{ per quarter} \times 4}{K11.29} = 1.9\%$$

As you can see, the current yield is actually shown in the stock table under Yld



Example

Find the PE ratio for each of the following corporations and round to the nearest whole number.

- Apple, Inc. (AAPL) with a closing price of $K116.52$ and annual earnings of $K8.31$
- Nike, Inc. (NKE) with a closing price of $K51.91$ and annual earnings of $K2.22$

Solution

- PE ratio for Apple, Inc. $= \frac{\text{Closing price per share}}{\text{Annual net income per share}} = \frac{K116.52}{K8.31} = 14$
- PE ratio for Nike, Inc. $= \frac{\text{Closing price per share}}{\text{Annual net income per share}} = \frac{K51.91}{K2.22} = 23$



Sinking Funds

- A **sinking fund** is a fund set up to receive periodic payments.
- Sinking funds are used to provide money to pay off a loan in one lump sum or to accumulate money to build new factories, buy equipment, and so on.
- The payment needed at the end of each period to accumulate a fixed amount in a sinking fund at a specific future date is found as follows:

$$\text{Payment} = \text{Future value} \times \text{Number from sinking fund table}$$



Example

In 5 years, hospital administrators plan to erect a building that includes a gym and a swimming pool that can be used for therapy, at a cost of $K16,500,000$. To prepare for the expense, they decide to make end-of-quarter deposits into a sinking fund expected to earn 6% compounded quarterly. Find

- i. the amount of each quarterly payment and
- ii. the interest earned.



Solution

- i. Use $\frac{6\%}{4} = 1.5\%$ per compounding period
- Number of compounding periods = 5×4 years = 20 compounding periods
 - In the sinking fund table 20 periods and 1.5% we get a value of 0.04325.

Quarterly payment = $K16,500,000 \times 0.04325 = K713,625$

Twenty end-of-quarter payments of $K713,625$ at 6% compounded quarterly will grow to $K16,501,629$.

- ii. Interest is the future value minus the payments.

Interest = $K16,501,629 - (20 \times K713,625) = K2,229,129$



Example

First Christian Church sold $K100,000$ worth of bonds that must be paid off in 8 years. It sets up a sinking fund to accumulate the necessary $K100,000$ to pay off the debt. Find the amount of each payment into a sinking fund if the payments are made at the end of each year and the fund earns 10% compounded annually. Find the amount of interest earned.



Solution

Look along the top of the sinking fund table for 10% and down the side for 8 periods to find 0.08744.

$$\text{Payment} = K100,000 \times 0.08744 = K8744$$

The church must deposit $K8744$ at the end of each year for 8 years into an account paying 10% compounded annually to accumulate $K100,000$. The interest earned is the future value less all payments.

$$\text{Interest} = K100,000 - (8 \times K8744) = K30,048$$



Setting up a sinking fund table

Example

First Christian Church in the previous Example deposited *K8744* at the end of each year for 8 years into a sinking fund that earned 10% compounded annually. Set up a sinking fund table for these deposits. After each calculation, round each answer to the nearest cent before proceeding.



Solution

- The sinking fund account contains no money until the end of the first year, when a single deposit of $K8744$ is made.
- Since the deposit is made at the end of the year, no interest is earned in the first year.
- At the end of the second year, the account contains the original $K8744$ plus the interest earned on the $K8744$.
- This interest is found by the formula for simple interest.

$$I = K8744 \times 0.10 \times 1 = K874.40$$



Solution continuation...

An additional deposit is also made at the end of the second year, so that the sinking fund then contains the following total.

$$K8744 + K874.40 + K8744 = K18,362.40$$

Continue to get the following sinking fund table.



	Beginning of Period		End of Period	
Period	Accumulated Amount	Periodic Deposit	Interest Earned	Accumulated Amount
1	K0	K8744.00	K0	K8744.00
2	K8744.00	K8744.00	K874.40	K18362.40
3	K18362.40	K8744.00	K1836.24	K28942.64
4	K28942.64	K8744.00	K2894.26	K40580.90
5	K40580.90	K8744.00	K4058.09	K53382.90
6	K53382.99	K8744.00	K5338.30	K67465.29
7	K67465.29	K8744.00	K6746.53	K82955.82
8	K82955.82	K8744.00	K8295.58	K100,000.00

The last payment is found as follows:

Desired Accumulated Amount - Interest Earned during Last Period - Accumulated Amount at end of Period 7

$$K100,000 - K8295.85 - K82,955.82$$



End of Lecture 6

